Technical Information

Requirements for the connection of electrically heated boilers

TI053
Version 3 (02/23)

1 General

This technical information describes the requirements for the connection of electrically heated boilers, and it should be used as an aid for the preparatory work to be carried out by the builder. The technical information also documents the interfaces, which are provided by Bosch within the scope of delivery, unless other agreements have been made on a specific order basis.

This document also applies equally to hybrid boilers, i.e. boilers that are operated with liquid and/or gaseous fuels and are electrically heated.

All relevant national and local regulations and applicable standards must be observed.

The term “Output Ptot (kW)” refers to the max. connected power at the power cabinet (and heating bundle). Fixed parameters have been defined for standardisation purposes. The actual output drawn is normally always smaller and varies from order to order. Among other things, it depends on the rated steam output, the operating pressure and temperature of the supplied feed-water/condensate.

To ensure the steam or hot water boiler is used as intended, the technical specifications on the order confirmation for the boiler system concerned must be observed.

2 Overview of the main components

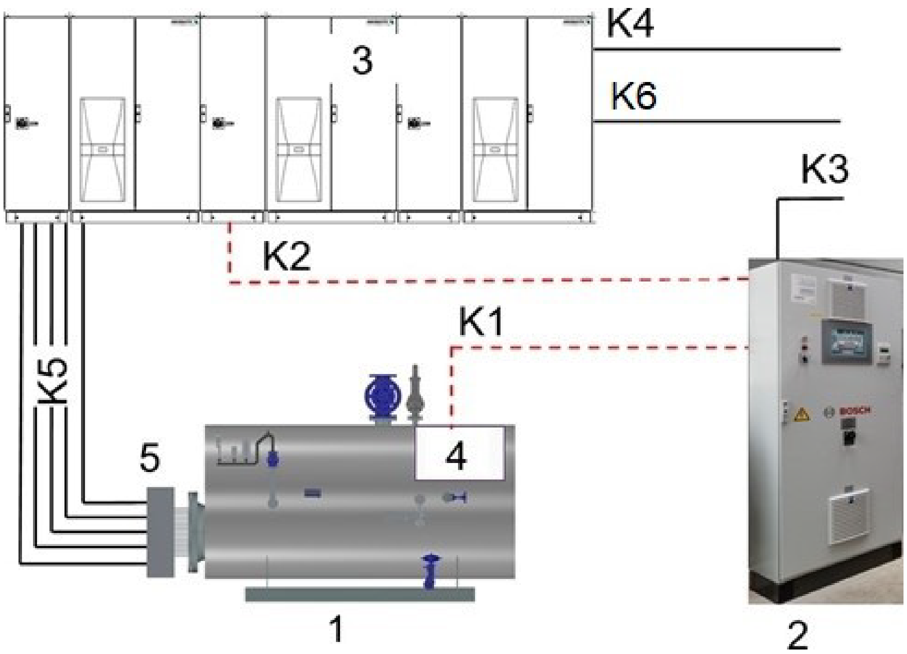

Fig. 2-1 Illustration of the connection between the main components

▬ ▬ ▬ ▬	Connection cable: optional Bosch	▬▬▬▬	Connection cable: provided by the builder
1	Boiler	K1	Connection of boiler control / boiler terminal box
2	Boiler control panel	K2	Control voltage of boiler control and interface between boiler control / power cabinet
3	Power cabinet	K3	Boiler control panel supply by the builder
4	Boiler terminal box	K4	Supply by the builder (incl. PE cables) to the power cabinet
5	Terminal box for heating bundle	K5	Power cable for switching stages
		K6	Supply to air-conditioning units (optional)

2.1 General

The minimum bending radiuses of the specified cable types must be observed and taken into account when planning the cable routing. Only approved and suitable cable routes and cable ducts are permitted, which are then also included in the protective equipotential bonding.

When laying cables, the customer must ensure that power, control and communication cables are routed separately in accordance with the guidelines in order to prevent interference (e.g. EMC).

In the case of communication cables, ensure that the shielding contacts have a sufficiently large surface area. Cable connections must be checked for tightness.

The maximum cable length (if specified) must be observed.

Protect the cables from direct sunlight.

2.1.1 Power cabinet

Features:

- Short circuit protected up to 50 kA, this must be taken into account, when fusing is provided by the builder. If necessary, measures must be taken by the builder to limit the short circuit current.
- Weight and main dimensions: H x W x D: 2200 mm x W x 600 mm. Dimension W depends on the output (dimension H incl. base) and the voltage.

Output	Control panel width W (mm)		Weight, approx. (kg)	
P_{ges} (kW)	400 V	690 V	400 V	690 V
250	1400	1400	700	750
500	2400	2400	850	900
1100	3800	3000	1750	1400
2200		3800		1750
3600		4800		2000
5500		7000		3250

Tab. 2-1 Width of power cabinet depending on the connected power P_{tot} and the supply voltage

The overall dimensions of the control cabinet must also allow for the following:
- H: + 70 mm for lifting lugs attached to the top of the power cabinet
- W: + 100 mm (max.) for protective cover of main switch
- W: + 300 mm if air-conditioning units (integrated in the doors) are used (option)

- Height of base: 200 mm
- Cabinet colour: RAL 7035.
- Internal subdivision according to IEC 61439-2:
  - ELSB1: Form 1
  - ELSB2 to ELSB6: Form 2b
- Degree of protection in accordance with EN60529: IP 54.
- Permissible voltage fluctuation: +/- 5 %.
- Permissible frequency fluctuation: +/- 1 %.
- Neutral conductor: no
- Earth conductor: yes
- Interior installation (no Ex protected zone)

- Mounting height: max. 2000 m (if higher, must be assessed on a case-by-case basis)
- Permitted ambient temperatures:
  - + 10 to +35 °C: standard version. With integrated temperature-controlled fans.
  - > 35°C to + 45°C: not included as standard and must be specified before ordering. With air-conditioning units integrated into the control cabinet doors. Separate 400 V power supply and connecting cable according to *Tab. 2-2 Number and connected power of air-conditioning units depending on connected power P_{tot}*.
  - > + 45°C: on request

Output	Air-conditioning units	
P_{ges} (kW)	Quantity	Connected power (kW) per air-conditioning unit
250	1	1.5
500	1	2.5
1100	2	1.5
2200	2	2.5
3600	2	2.5
5500	3	2.5

Tab. 2-2 Number and connected power of air-conditioning units depending on connected power P_{tot}

- Deliveries are split as follows:
  - Outputs ≤ 3600 kW, assembled in one piece.
  - Output 5500 kW in 2 parts for assembly on site. (Part 1: W= 4800; Part 2 W = 2200 mm)

  For safety reasons, disassembly into smaller units, e.g. for inward transportation, is not permitted.
- Feed supply terminal clamps to the connection rail of the power cabinet are not included in the scope of delivery.
- The power cabinet supply cable is delivered by the builder.

2.1.2 Heating bundle (in the boiler) with attached terminal box

- Control panel design: stainless steel 1.4301
- Degree of protection in accordance with EN60529: IP 54.
- Interior installation (no Ex protected zone)
- Attached to the heating bundle (which is installed in the boiler) when delivered.
- Delivery of the connection cables between the power cabinet and the heating bundle terminal box to be provided by the builder.

2.1.3 Boiler terminal box (attached to the boiler)

- Attached to the boiler when delivered (as long as the boiler control is supplied by Bosch).
- The wiring of the sensors and actuators, which are attached to the boiler, with the boiler terminal box is included in the scope of delivery by Bosch.

2.1.4 Boiler control panel

- Short circuit protected up to 10 kA (given on the "Icu" rating plate). This must be taken into account, when fusing is provided by the builder. If necessary, measures must be taken by the builder to limit the short circuit current.
- Cabinet colour: RAL 7035.
- Degree of protection in accordance with EN60529: IP 54.
- Permissible voltage fluctuation: +/- 5 %.
- Permissible frequency fluctuation: +/- 1 %.
- Neutral conductor: yes
- Earth conductor: yes
- Cabinet design:

  - Solely electrically operated boilers: Wall cabinet (size as required). Optional free-standing cabinet: Dimensions; H x W x D: 2200 mm x W x 600 mm. Width as required (height including base)
  - Hybrid boiler (boiler with fossil heating and additional electric heating): Dimensions of free-standing cabinet; H x W x D: 2000 mm x W x 400 mm. Width as required (height including base)
- Interior installation (no Ex protected zone)
- Connection cable between boiler control cabinet and boiler terminal box can be supplied by Bosch if required (see offer / order confirmation).
- Connection cable (control lines) between the boiler control cabinet and the power cabinets can be supplied by Bosch if required (see offer / order confirmation).
- Connection cable (control lines and power cable) between the boiler control panel and actuators/sensors installed in the boiler house (e.g. pumps) can be supplied by Bosch if required (see offer / order confirmation).
- The supply cable to the boiler control panel is delivered by the builder (note: always 400 V +/- 5 %).

2.2 Feed supply to power cabinet

2.2.1 Connected load ≤ 3600 kW

A central supply point is provided in the power cabinet. These are copper rails, to which bolts are attached. The cables provided by the builder can be attached to this supply point. These are incorporated from below via the base (base height: 200 mm) in the first field of the power cabinet (from the left, when standing in front of the power cabinet).

As a basic rule, a clockwise rotating field must be set up at the supply point of the power cabinet for the main switch.

This must be carried out by the system installer (through their qualified electricians) and documented in writing.

2.2.2 Connected load 5500 kW

Two feed points are provided in the power cabinet. Both are copper rails on which bolts are mounted. The on-site cables can be connected to these.

Feed point 1: Integration of 3600 kW, is done from below via the base (base height: 200 mm) in the 1st field of the power cabinet (from the left, standing in front of the power cabinet).

Feed point 2: Integration of 1900 kW, is done from below via the base (base height: 200 mm) in the 7th bay of the power cabinet (from the left, standing in front of the power cabinet).

Basically, a clockwise rotating field must be created at both feed points of the power cabinet for both main switches.

This must be carried out by the system installer (through their qualified electricians) and documented in writing.

2.2.3 For all connection loads

The incoming feeder cable must be sized on-site. The sizing of the feed supply line must be carried out in accordance with the applicable local regulations, and it must be ensured that the maximum short circuit current for the control panels is not exceeded, and that the ambient conditions are taken into account, as well as the required length of the supply line when sizing. That is to say, the connection in the power cabinet is made via a cable-connected feed supply (no copper rails).

When installing the switchgear, sufficient space must be provided under the interception rail for the bending radius of the connection cables.

Depending on the size of the boiler, the cable connections given in Table 2-2 are provided in the power cabinet for the feed supply. If there are any discrepancies, agreement with Bosch is required as mandatory, so that any problems with connecting the feed supply are avoided.

The requirements included with the order-related wiring diagram must be noted, when finally completing the connection of the feed supply.

Output	Voltage	Supply		Load cable		PE cable		Feeding point
P_{ges} (kW)	U_N (V)	I_N (A)	I_N + 10 % (A)	Quantity	Cross-section (mm²)	Quantity	Cross-section (mm²)	
250	**400**	360.8	**396.9**	2	**4 x 120**	--	--	1
500	**400**	721.7	**793.9**	4	**4 x 150**	--	--	1
1100	**400**	1587.7	**1746.5**	21	**1 x 240**	4	**1 x 240**	1
250	**690**	209.2	**230.1**	1	**4 x 150**	--	--	1
500	**690**	418.4	**460.2**	2	**4 x 185**	--	--	1
1100	**690**	920.4	**1012.5**	12	**1 x 240**	2	**1 x 240**	1
2200	**690**	1840.8	**2024.9**	24	**1 x 240**	4	**1 x 240**	1
3600	**690**	3012.3	**3313.5**	39	**1 x 240**	7	**1 x 240**	1
5500	**690**	3012.3	**3313.5**	39	**1 x 240**	7	**1 x 240**	1
		1589.8	**1748.8**	24	**1 x 240**	4	**1 x 240**	2

Tab. 2-3 Designated supply connections at the power cabinet

For the 5500 kW output, the total output is divided between 2 feed points.

For the outputs 250 and 500 kW, it is envisaged that the connection is established using multi-core cables. The PE conductor is integrated into these cables.

Individual cables are planned for outputs 1100 to 5500 kW. Separate PE cables must be installed.

Minimum requirements for the cables:

- Cables for fixed installation with the type designation "NYY" are used.
- Cables with insulation and a casing made of thermoplastic PVC with rated voltage of 0.6/1 kV.
- Permissible operating temperature at the conductor: 70 °C.

2.3 Cable connection between the power cabinet and the heating bundle terminal box

2.3.1 Power cable

Basic principles for calculations and design are provided by EN50565. The following preconditions were taken as the basis for the cable cross-sections given in Table 3:

- The power cabinet and boiler with built-in heating bundle are located in one or more rooms with the same ambient conditions.
- Permitted ambient temperatures (room temperature) between +5 °C and +35 °C.
- Maximum line length between heating bundle terminal box and power cabinet is 20 metres.
- Cables with the "JZ-600" design abbreviation are used for flexible installation.
- Cables with insulation and a casing made of thermoplastic PVC with rated voltage of 0.6/1 kV.
- Permissible operating temperature at the conductor: 80 °C.
- Cable routing on perforated cable tray (horizontal or vertical). Dimensioning and installation must be carried out according to the cable tray manufacturer's specifications.
- Provide on-site measures (e.g. support beams) to ensure that no mechanical load is exerted on the terminal box by the connecting cables.

The power cable exits from below via the base (base height: 200 mm) of the power cabinet, and it is also fed from below into the heating bundle terminal box. If the output is ≥ 2,200 kW, additional space (min. + 200 mm) should be created by the builder underneath the power cabinet, so that the connection cables can be arranged (e.g. by means of a floor channel or additional base provided by the builder).

Depending on the output of the boiler, the cable connections given in Table 2-3 are provided in the heating bundle terminal box. If there are any discrepancies, agreement with Bosch is required as mandatory, so that any problems with connecting the feed supply are avoided.

The requirements included with the order-related wiring diagram must be noted, when finally completing the connection of the feed supply.

Output	Voltage	Increments		Feed supply		Cable		
P_{ges} (kW)	U_N (V)	Design	number	P_N (A)	I_N + 10 % (A)	number	Cross-section (mm²)	Cable trays *)
250	**400**	Thyristor	1	250	396,9	2	**4 x 120**	1
500	**400**	Thyristor	1	200	317,5	2	**4 x 95**	1
		Contactor	3	100	158,8	3	**4 x 95**	1
1100	**400**	Thyristor	1	183,4	291,2	2	**4 x 95**	1
		Contactor	10	91,7	145,6	10	**4 x 95**	1
250	**690**	Thyristor	1	250	230,1	1	**4 x 150**	1
500	**690**	Thyristor	1	200	184,1	1	**4 x 120**	1
		Contactor	3	100	92,0	3	**4 x 50**	1
1100	**690**	Thyristor	1	366,8	337,6	2	**4 x 95**	1
		Contactor	4	183,4	168,8	4	**4 x 95**	1
2200	**690**	Thyristor	1	400	368,2	2	**4 x 120**	2
		Contactor	9	200	184,1	9	**4 x 120**	2
3600	**690**	Thyristor	1	422,2	388,6	2	**4 x 150**	2
		Contactor	15	211,8	194,9	15	**4 x 150**	2
5500	**690**	Thyristor	1	423	389,3	2	**4 x 150**	3
		Contactor	24	211,5	194,7	24	**4 x 150**	3

Tab. 2-4 Designated supply connections at the terminal box for the heating bundle

*) the specified number of cable trays for the cable routing between the power screw and the heating bundle terminal box is based on the max. permissible bundling of cables per cable tray. The terminal box is connected with multi-core cables.

2.3.2 Control cable

The safety temperature limiter installed in the heating bundle is to be wired between the power cabinet and the heating bundle terminal box, using the cable design defined below.

- Type LiYCY 7x1.5mm²

The requirements included with the order-related wiring diagram must be noted, when finally completing the connection of the feed supply.

It is mandatory to ensure, that the control line is laid separately from the power cables, so that any issues with electromagnetic compatibility are prevented.

3 Earthing and potential equalisation

Earthing and potential equalisation must be carried out by the builder in accordance with the "Requirements for protection against electric shock" in IEC 60364-4-41 (or in Germany DIN VDE 0100-410).

The technical design of the potential equalisation and the dimensions of the cross-sections must be in accordance with IEC 60364-5-54 (or according to DIN VDE 0100-540 in Germany)

The implementation involves for example piping, flanges, valves, measuring equipment, motors, pumps, boiler pressure vessels, boiler and system components and control panels etc. In the case of boilers or boiler and system components, earthing can generally be carried out at the base frame (e.g. via earthing terminals). The connection between the base frame and the foot of the boiler or tank must be made via a conductive connection and connected to the same earth potential.

The design of this must be in accordance with the regulations that apply locally, and where applicable the manufacturer's instructions for the individual components must be observed.

The minimum requirements are:

- It must be ensured that the earthing measures are mechanically secure and corrosion-resistant.
- Provision must be made for the highest fault current (as calculated) from the thermal point of view.
- Damage to units, components and other operating equipment must be prevented.
- The safety of persons must be ensured as regards voltage on earthing systems arising from the highest possible fault current.
- The positions, which are used for potential equalisation, must be ground bare (removal of anti-corrosion paint) and provided with a suitable marking.
- After potential equalisation has been completed, the areas that have been ground bare must be protected once more against corrosion.

4 Important information about load shedding

The load control for the electrically operated boiler is preset by Bosch in such a way, that the load range from 0 to 100 % can be ramped up or down in a **minimum** of 120 seconds.

Despite all these measures, a safety-related shutdown can sometimes occur due to the boiler's safety chain responding (or in the power cabinet). This can lead to the boiler system shutting down abruptly. This can not be foreseen and can occur at any load setting, even at 100% load in an extreme case.

The maximum output P_{ges} (kW) applicable to your boiler can be found in Table 2-1 or in the quotation/order confirmation.

The operator must clarify with the local energy supply company, how an instance of abrupt load shedding from 100% load (in the event of a safety-related shutdown) is to be handled, and which measures need to be taken by the builder, so that no negative effects on the supply network are created by an abrupt shutdown at full load.

5 Liability

If the above points and the requirements given in the corresponding operating instructions are not followed, damage may be caused to the boiler and boiler components, as the boiler manufacturer, will accept no liability in such cases.